ScienceX

Battery Storage Market Notes, October 2026 issue

Science X Inc. Published November 1, 2026

The price of electricity changes with the time of day it is used. "Grid-scale battery storage" (large batteries connected to the transmission grid that store electricity and release it) earns money from that difference. These notes use published figures and our own records to show this month's movements in figures and tables.

1. The three income streams of a battery storage plant (how to read these notes)

A battery storage plant (grid-scale battery storage) earns income from three markets. ① Wholesale electricity (JEPX) is the market where you buy and store electricity when it is cheap and sell it when it is expensive. ② The balancing market (EPRX) is the market where you sell grid operators the "capacity" to correct surpluses or shortages of electricity. ③ The capacity market (OCCTO) is the market where the government buys future electricity supply capacity in advance. These notes set out the numbers for the three markets in this order. The ①, ② or ③ at the start of each section title shows which market it covers.

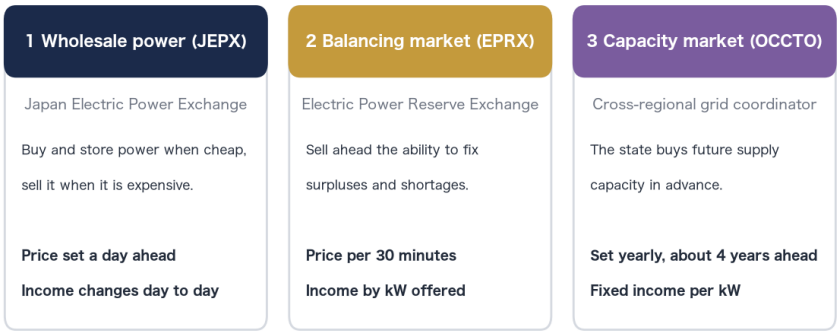


Figure 1 The three income streams of a battery storage plant

Source: Rules of the Japan Electric Power Exchange (JEPX), the Electric Power Reserve Exchange (EPRX) and the Organization for Cross-regional Coordination of Transmission Operators (OCCTO).

2. Three movements this month

We picked three movements this month from the numbers of the three markets.

- In wholesale electricity (JEPX), the monthly average gap between the most expensive 4 hours and the cheapest 4 hours in October 2026 was 17.14 yen/kWh in Tokyo and 19.30 yen/kWh in Kyushu.
- In the balancing market (EPRX), the bid ratio for Primary reserve passed 100% for the first time since April 2025, at 106.8% in June 2026. In September 2026 it was 123.1%, and the average winning price was 3.43 yen/kW per 30 min.
- In the balancing market (EPRX), 56.9% of Tertiary reserve 2 revenue in September 2026 was concentrated in the top 10% of 30-minute slots by revenue.

3. ① Wholesale electricity (JEPX) | Average price

Wholesale electricity is the electricity that power companies and retail suppliers buy and sell among themselves. Its price is set on the "spot market" of the Japan Electric Power Exchange (JEPX), where electricity for the next day is traded the day before. "System (nationwide)" is the price when all of Japan is treated as one market (the system price). When the transmission lines between regions are congested, the price splits into separate prices for each area from Hokkaido to Kyushu (area prices).

The average System (nationwide) price for wholesale electricity in October 2026 was 22.51 yen/kWh, compared with 11.67 yen/kWh in the same month last year.

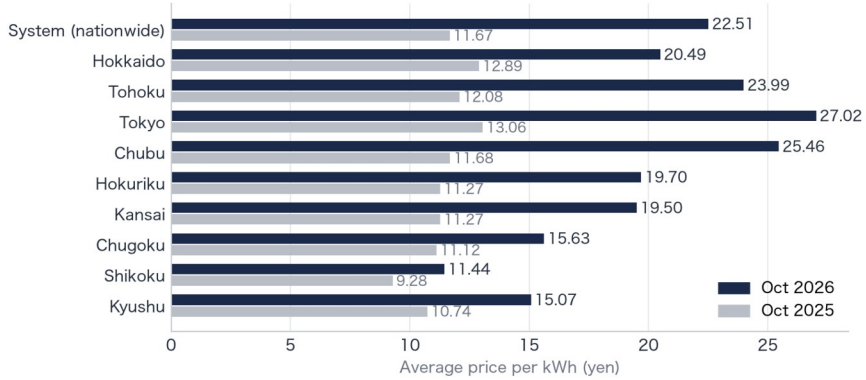


Figure 2 Average wholesale electricity price (yen/kWh). Navy is October 2026, grey is the same month last year

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Simple average of the 30-minute prices. Our own tally.

4. ① Wholesale electricity (JEPX) | How the price moves through the day

Wholesale electricity prices change every 30 minutes. The figure shows the shape of an average day: the prices at the same time of day are averaged across the month. If a battery stores electricity in the cheap hours and releases it in the expensive hours, the difference becomes its income.

The daily average low and high for wholesale electricity were 19.03 yen around 12:00 and 37.11 yen around 16:30 in Tokyo, and 4.94 yen around 12:00 and 24.74 yen around 17:30 in Kyushu (yen/kWh).

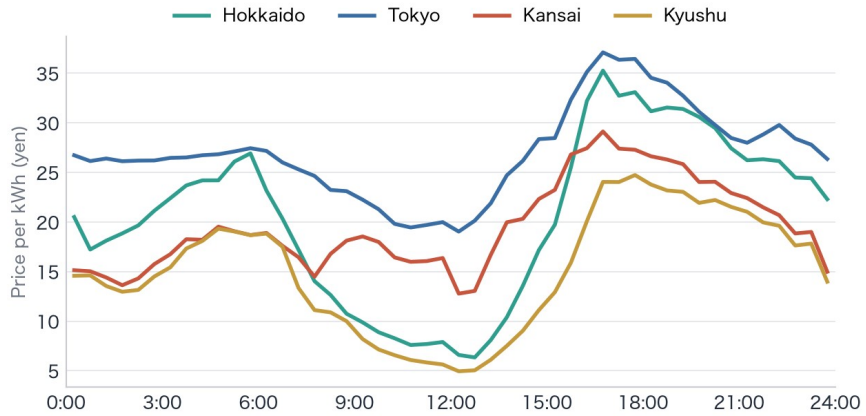


Figure 3 Daily peaks and troughs of wholesale electricity prices (October 2026 average, yen/kWh)

Source: JEPX spot market. Our own tally.

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5. ① Wholesale electricity (JEPX) | How large is the price gap per kWh

For each area, we took the average price of the day's most expensive 4 hours, subtracted the average price of the day's cheapest 4 hours, and show the monthly average. This equals the gain per kWh from buying 1 kWh in the cheapest 4 hours and selling it in the most expensive 4 hours. It is a simple difference. It does not deduct electricity lost in charging and discharging or fees, and it ignores whether the cheap hours come before or after the expensive hours.

By area, the monthly average gap between the most expensive and the cheapest 4 hours of wholesale electricity was largest in Hokkaido at 29.02 yen/kWh and smallest in Tokyo at 17.14 yen/kWh.

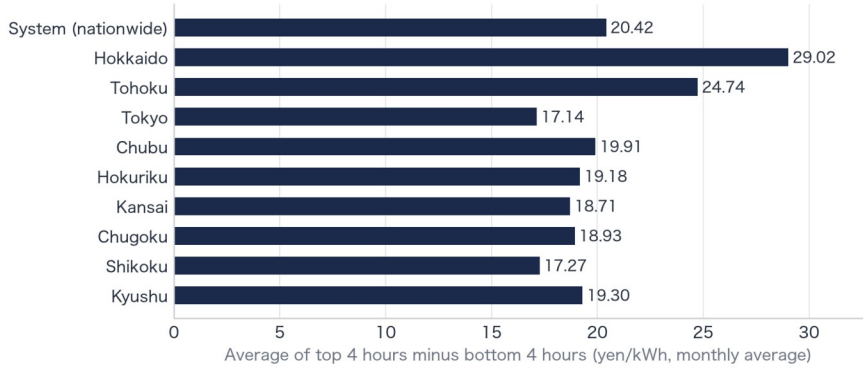


Figure 4 Wholesale electricity price gap: average of the most expensive 4 hours − average of the cheapest 4 hours (October 2026, yen/kWh, monthly average)

Source: JEPX spot market. Our own tally.

Area	2-hour gap (yen/kWh)	4-hour gap (yen/kWh)	Days with a 4-hour gap of 10 yen or more
System (nationwide)	22.95	20.42	29
Hokkaido	32.56	29.02	31
Tohoku	27.09	24.74	30
Tokyo	19.62	17.14	29
Chubu	22.37	19.91	29
Hokuriku	21.71	19.18	31
Kansai	21.22	18.71	31
Chugoku	20.90	18.93	31
Shikoku	19.40	17.27	28
Kyushu	21.19	19.30	31

The gaps are monthly averages of the daily gaps. October 2026 covers 31 days.

6. ① Wholesale electricity (JEPX) | There is a price gap every day

We show the same 4-hour gap day by day. The bigger the gap on a given day, the more room a battery has to earn from the price gap.

The number of days with a 4-hour wholesale price gap of 10 yen or more was 29 of 31 for System (nationwide), 31 in Hokkaido, Hokuriku, Kansai, Chugoku and Kyushu, and 28 in Shikoku.

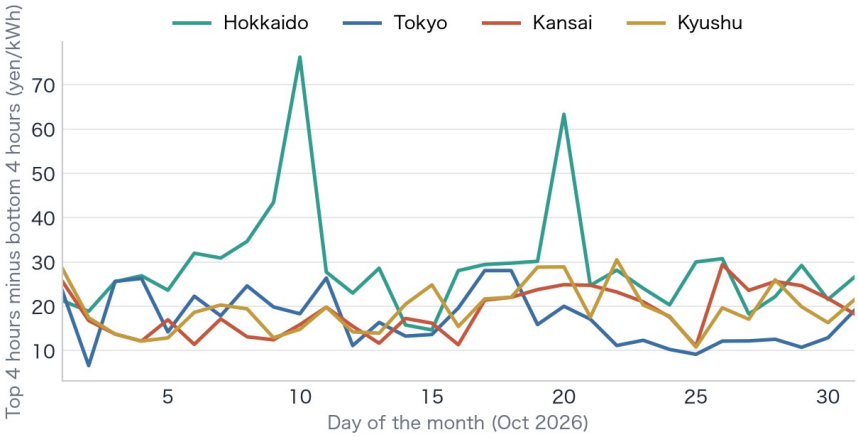


Figure 5 Daily wholesale electricity price gap (average of the day's most expensive 4 hours − average of the cheapest 4 hours, yen/kWh)

Source: JEPX spot market. Our own tally.

7. ② Balancing market (EPRX) | Six products, and prices by area

The balancing market is where grid operators buy, every 30 minutes, the "capacity" to correct surpluses or shortages of electricity. The products differ by how fast output must change after an instruction. Primary reserve is the fastest, and Tertiary reserve 2 is the slowest. A Composite product is a bid that offers several products together. Prices are set in yen per 1 kW of capacity offered per 30 minutes. The figure shows the range of winning prices by area (by the area where the power source is located). The thin line runs from the lowest to the highest winning bid in the month, the thick band is the price range that 8 in 10 slots with winning bids fall into, and the dot is the average winning price. Nationwide figures are in the tables in the "Market figures" section. Published figures for October 2026 are not yet in our tally, so we show the latest month with all products available: September 2026 (1st to 29th), preliminary figures.

In September 2026, by the area where the power source is located, the highest average winning price in the balancing market was Primary reserve in Tohoku at 8.18 yen, and the lowest was Tertiary reserve 2 in Shikoku at 0.38 yen (yen/kW per 30 min).

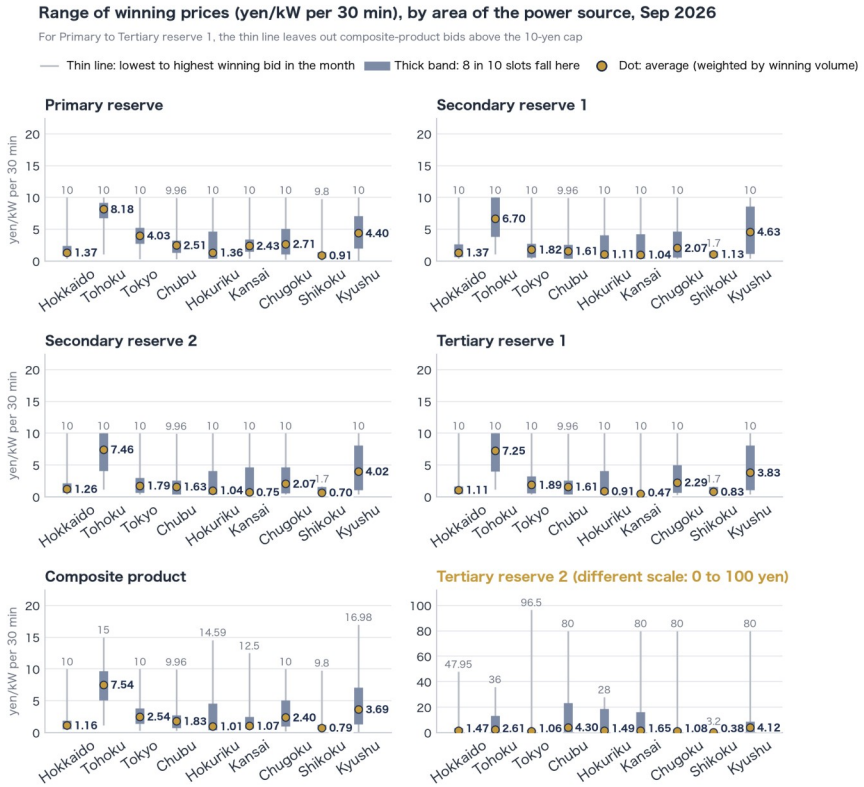


Figure 6 Range and average of winning prices by area in the balancing market (September 2026 (1st to 29th), by the area where the power source is located)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own tally.

Product	From instruction to change in output	Duration
Primary reserve	Within 10 seconds	5 minutes or more
Secondary reserve 1	Within 5 minutes	30 minutes
Secondary reserve 2	Within 5 minutes	30 minutes
Tertiary reserve 1	Within 15 minutes	30 minutes
Tertiary reserve 2	Within 60 minutes	30 minutes
Composite product	Requirements of the combined products	Requirements of the combined products

8. ② Balancing market (EPRX) | Where and when bids win

The clearing rate is the share of bid volume that won, and the higher it is, the more easily a bid wins. The figure shows, for each grid area (the area where the power source is located), the clearing rate of each product by time of day (every 30 minutes). Where a line breaks, there were no bids at that time.

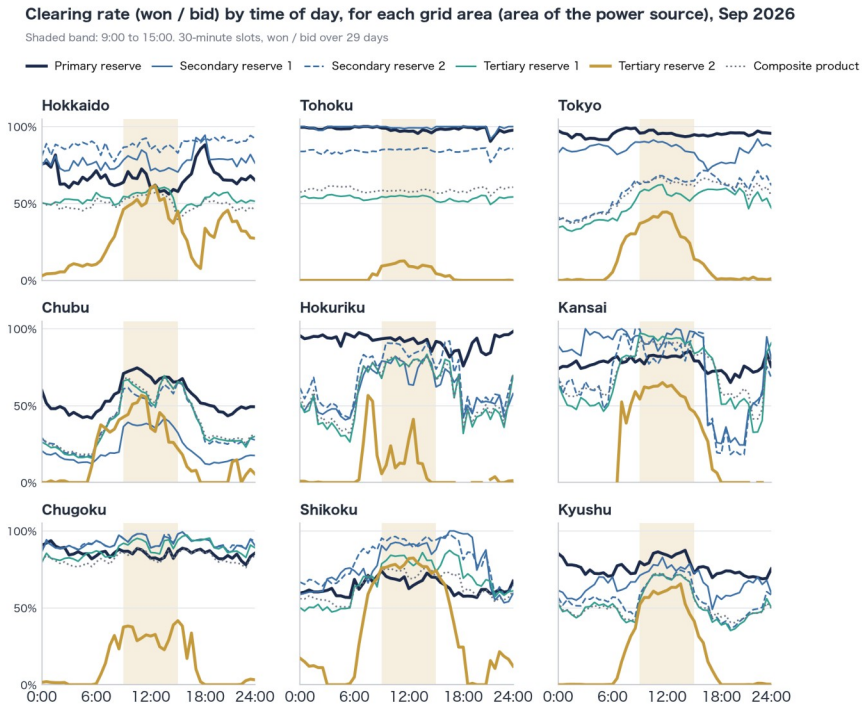


Figure 7 Clearing rate in the balancing market (grid area (the area where the power source is located) × time of day, September 2026 (1st to 29th))

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own tally.

9. ② Balancing market (EPRX) | Primary reserve has become a race for seats

Primary reserve is the product where batteries have the highest share of the winning volume. The bars show the bid ratio (bid volume ÷ volume requested by grid operators). Above 100%, more bids come in than the volume requested, and some bids lose. The line shows the average winning price. These are the monthly figures we have, starting from April 2025.

The bid ratio for Primary reserve in the balancing market passed 100% for the first time since April 2025 in June 2026 (106.8%), and was 123.1% in September. The average winning price was 3.43 yen/kW per 30 min, compared with 3.57 yen in the same month last year.

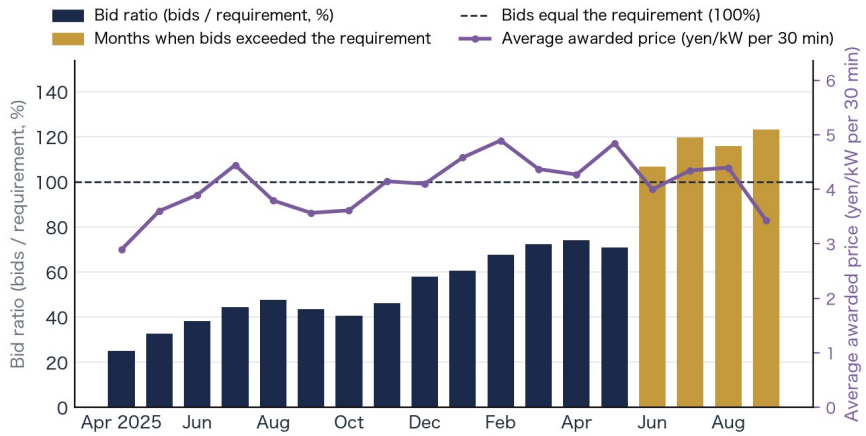


Figure 8 Primary reserve bid ratio (bars) and average winning price (line) (monthly; gold bars are months when bids exceeded the volume requested)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own tally.

10. ② Balancing market (EPRX) | How prices have moved for each product

This is the average winning price for each product since April 2025 (monthly average weighted by winning volume). From March 14, 2026, Primary reserve to Tertiary reserve 1 and the Composite product switched from trading in 3-hour blocks to trading every 30 minutes (Tertiary reserve 2 was already traded every 30 minutes).

The average winning price for Primary reserve in the balancing market was 2.89 yen in April 2025 and 3.43 yen in September 2026, with a high of 4.90 yen in February 2026 (yen/kW per 30 min).

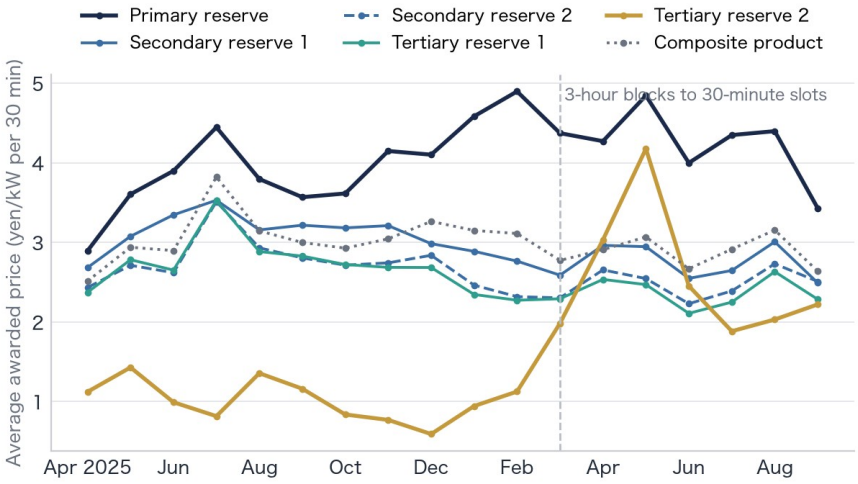


Figure 9 Average winning price by product in the balancing market (yen/kW per 30 min, monthly)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own tally.

11. ② Balancing market (EPRX) | Which hours of the day are expensive

This is the average winning price by hour for each product in September 2026 (1st to 29th).

In September 2026, the average winning price for Primary reserve in the balancing market was highest around 18:00 at 4.09 yen and lowest around 00:00 at 2.75 yen (yen/kW per 30 min).

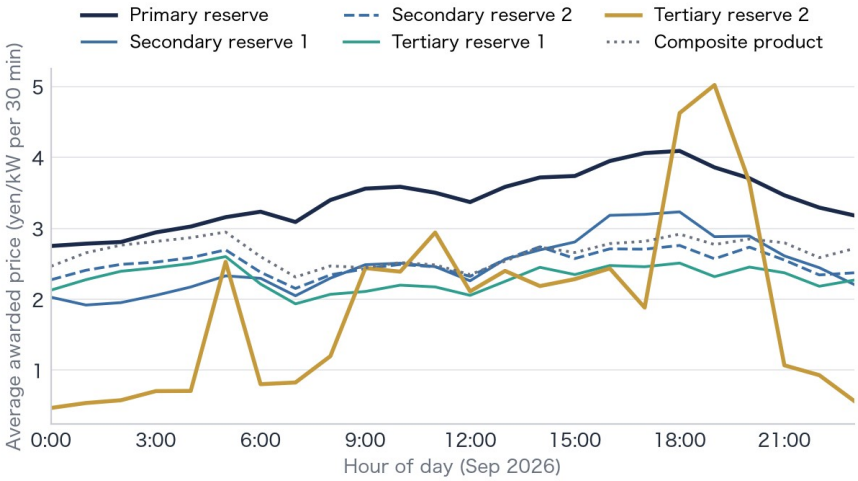


Figure 10 Average winning price by time of day in the balancing market (September 2026 (1st to 29th), yen/kW per 30 min)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own tally.

12. ② Balancing market (EPRX) | Which products batteries are used in

The figure shows the share of winning volume taken by batteries. The table compares the average winning price of the volume won by batteries with the average winning price of the product as a whole.

For Primary reserve in the balancing market (September 2026), the battery share was 27.9%. The average winning price of the volume won by batteries was 5.17 yen/kW per 30 min, compared with 3.43 yen overall.

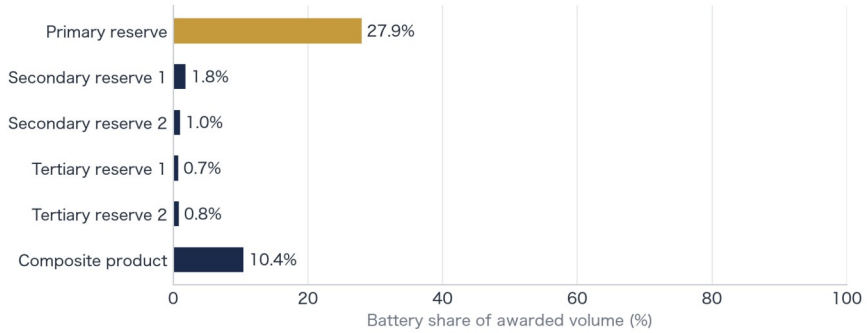


Figure 11 Share of batteries in the balancing market (%)

Source: Trading results by power source type (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own tally.

Product	Won by batteries (yen/kW per 30 min)	Product overall (yen/kW per 30 min)
Primary reserve	5.17	3.43
Secondary reserve 1	2.18	2.49
Secondary reserve 2	1.85	2.51
Tertiary reserve 1	1.86	2.28
Tertiary reserve 2	59.26	2.22
Composite product	5.13	2.64

Both are averages weighted by winning volume (September 2026, preliminary, nationwide). For Tertiary reserve 2, batteries account for only 0.8% of the winning volume, and most of it was won in high-price slots, so the average comes out high.

13. ② Balancing market (EPRX) | Revenue is concentrated in a small number of slots

For each product, we ranked the month's 30-minute slots that had winning bids by revenue (average winning price × winning volume, nationwide), from highest to lowest, and plotted what % of the month's revenue the top slots account for. The more the curve bulges above the dotted line, the more the revenue is concentrated in a few slots. Tertiary reserve 2 is a product that covers errors in forecasting renewable output caused by the weather. Its price is low in most slots and high in a very small number of slots.

In the balancing market (September 2026), the share of revenue concentrated in the top 10% of slots was 56.9% for Tertiary reserve 2 and 15.8% for Primary reserve.

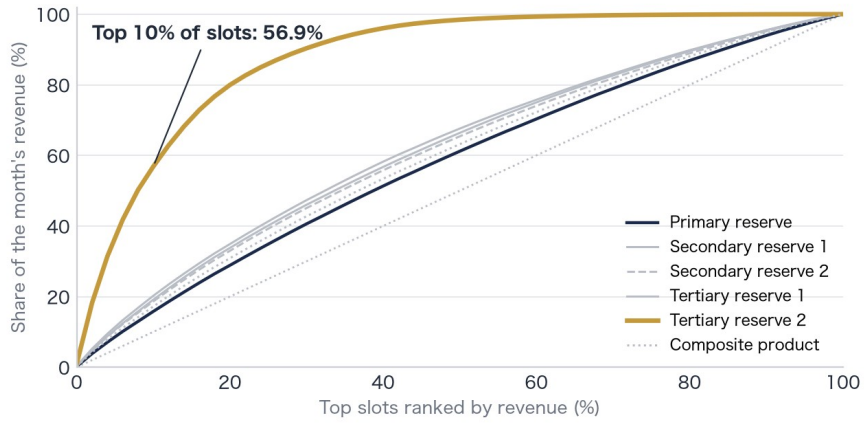


Figure 12 Share of balancing market revenue concentrated in the top slots (September 2026 (1st to 29th), nationwide)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own tally.

In September 2026, out of the 1,194 slots with winning bids, the top 1% of slots by revenue (average winning price × winning volume) accounted for 9.4% of the month's revenue. There were 25 slots where the nationwide average winning price was 10 yen/kW per 30 min or more, and 198 slots with no winning bids.

Slots with the highest maximum winning prices (by the area where the power source is located):

Date	Time	Area	Maximum winning price (yen/kW per 30 min)
September 8	09:30	Tokyo	96.50
September 8	10:30	Tokyo	96.50
September 8	11:00	Tokyo	96.50
September 27	11:00	Tokyo	96.00
September 27	11:30	Tokyo	96.00

Source: Tertiary reserve 2 trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own tally.

14. ③ Capacity market (OCCTO) | A market that buys future supply capacity

The capacity market buys electricity supply capacity four years ahead through a once-a-year auction. The price is set in yen per kW per year. The left figure shows the clearing prices of past main auctions. The right figure shows the battery and pumped hydro capacity that cleared in the Long-Term Decarbonization Power Source Auction (a separate auction that supports new builds of batteries, pumped hydro and others). These figures are published once a year, so they do not change from month to month. The latest is for delivery FY2029 (published January 2026).

The capacity market clearing price in Tokyo went from 3,495 yen/kW for delivery FY2025 to 15,111 yen/kW for FY2029. In the Long-Term Decarbonization Power Source Auction, batteries and pumped hydro bid in FY2025 cleared 819 MW of lithium-ion and 886 MW of non-lithium. Both are the latest once-a-year published figures.

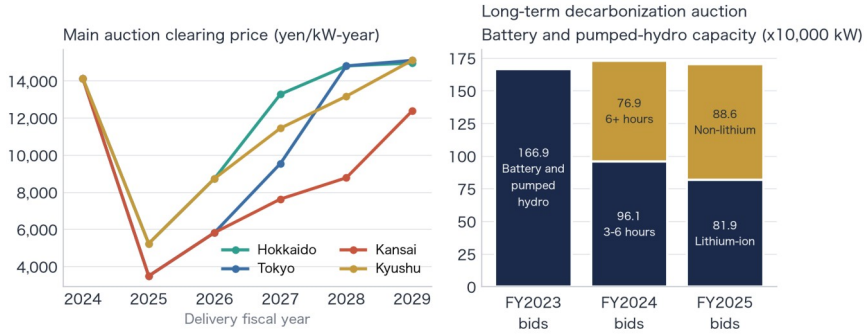


Figure 13 Capacity market clearing prices (left, yen/kW) and cleared battery and pumped hydro capacity in the Long-Term Decarbonization Power Source Auction (right, in units of 10,000 kW)

Source: Clearing results and capacity assurance contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in section 15). As published.

15. Our records | Where battery storage projects are increasing

We counted the projects newly listed this month on DealRoom (our data room for buyers, a site where we list battery storage projects for buyers), by grid area of the grid operator. These are numbers for the projects we handled, not counts for the whole market. Transfer of rights is a way of selling by handing over the rights to the land and the power. Completed-build handover is a way of selling by handing over the facility once it is built.

In October 2026, DealRoom had 77 new listings totalling 309.1 MW. By grid area, Tokyo had 33 projects and 99.8 MW, Tohoku 8 projects and 73.4 MW, and Chubu 11 projects and 69.9 MW.

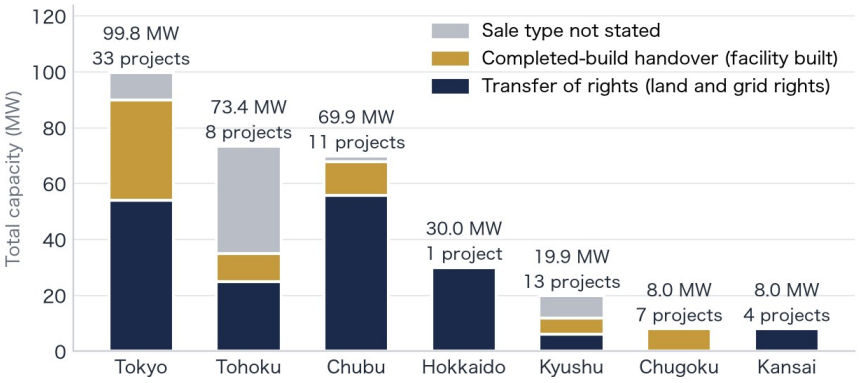


Figure 14 Output of the projects we listed on DealRoom in October 2026 (MW)

Source: Our DealRoom records (project locations are given only down to the grid area; prices and parties are not listed).

16. Our records | How many months and how much it takes to connect to the grid

A grid connection study is a procedure where you ask the grid operator to check "Can we connect at this location, and how much will the work cost?" The grid connection cost contribution is the part of the connection work cost paid by the applicant. From the responses we received, we tallied only the high-voltage cases (connections to 6,600 V distribution lines), by grid area. Extra-high voltage is not included. Grid areas with fewer than 5 cases are not shown.

The medians of the grid connection study responses (high voltage only, as of October 19, 2026) ranged from 8.0 months in Tokyo to 17.0 months in Tohoku for the construction period, and from ¥3.02 million in Tokyo to ¥32.35 million in Tohoku for the grid connection cost contribution.

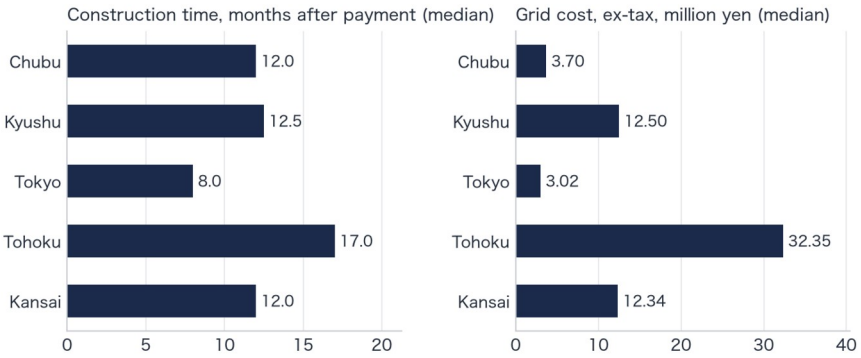


Figure 15 Median grid connection study responses (high voltage) by grid area (left: months of construction work, right: grid connection cost contribution, excluding tax)

Source: Tally of the high-voltage grid connection study responses we received from general transmission and distribution operators (as of October 19, 2026).

17. Market figures (tables ready to use in your materials)

We have put the numbers behind the figures in the main text into tables you can use in your materials: ① wholesale electricity (JEPX) figures by area and trading volume, ② nationwide figures for each balancing market product (bid ratio, clearing rate, price, battery share), and ③ capacity market clearing prices.

Wholesale electricity, JEPX spot market (October 2026, by area, yen/kWh)

Area	Monthly average	Same month last year	Daytime (08:00–16:00)	Evening (16:00–20:00)	4-hour gap	Maximum	30-minute slots at 30 yen or more	30-minute slots at 0.01 yen
System (nationwide)	22.51	11.67	16.02	31.06	20.42	47.88	220	39
Hokkaido	20.49	12.89	11.31	32.25	29.02	105.00	303	67
Tohoku	23.99	12.08	15.78	34.19	24.74	50.01	437	95
Tokyo	27.02	13.06	23.12	34.69	17.14	50.01	466	16
Chubu	25.46	11.68	20.38	33.55	19.91	46.45	387	24
Hokuriku	19.70	11.27	18.29	26.92	19.18	46.45	218	24
Kansai	19.50	11.27	18.22	26.76	18.71	46.45	215	29
Chugoku	15.63	11.12	9.86	23.10	18.93	44.79	63	79
Shikoku	11.44	9.28	7.50	16.61	17.27	37.51	23	146
Kyushu	15.07	10.74	8.31	23.10	19.30	44.79	53	91

Trading volume (October 2026, total of 31 days): total cleared volume 30,319 million kWh (same month last year 22,279 million kWh), sell bid volume 45,218 million kWh, buy bid volume 38,191 million kWh.

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Prices are simple averages of the 30-minute slots. Our own tally.

Balancing market (September 2026, preliminary, nationwide)

Product	Volume requested (MW)	Bid volume (MW)	Winning volume (MW)	Bid ratio (bids ÷ requested)	Clearing rate (won ÷ bids)	Average winning price (yen/kW per 30 min)	Battery share
Primary reserve	1,388.7	1,710.0	1,337.3	123.1%	78.2%	3.43	27.9%
Secondary reserve 1	962.6	1,996.2	1,102.3	207.4%	55.2%	2.49	1.8%
Secondary reserve 2	983.1	3,709.3	2,163.2	377.3%	58.3%	2.51	1.0%
Tertiary reserve 1	2,498.5	5,895.9	3,021.0	236.0%	51.2%	2.28	0.7%
Tertiary reserve 2	376.3	1,371.7	368.9	364.5%	26.9%	2.22	0.8%
Composite product	3,383.8	6,641.4	3,623.5	196.3%	54.6%	2.64	10.4%

Volumes are averages per 30-minute slot. A product with a bid ratio below 100% does not get enough bids for the volume requested. The clearing rate is the share of bid volume that won.

Source: Trading results (preliminary) and trading results by power source type published by the Electric Power Reserve Exchange (EPRX). Our own tally.

Capacity market main auction area prices (yen/kW per year)

Delivery fiscal year	Hokkaido	Tohoku	Tokyo	Chubu	Hokuriku	Kansai	Chugoku	Shikoku	Kyushu
FY2024	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137
FY2025	5,242	3,495	3,495	3,495	3,495	3,495	3,495	3,495	5,242
FY2026	8,749	5,833	5,834	5,832	5,832	5,832	5,832	5,832	8,748
FY2027	13,287	9,044	9,555	7,823	7,638	7,638	7,638	7,638	11,457
FY2028	14,812	14,812	14,812	10,280	8,785	8,785	8,785	8,785	13,177
FY2029	14,972	15,111	15,111	12,388	12,388	12,388	12,388	12,388	15,112

Cleared battery and pumped hydro capacity in the Long-Term Decarbonization Power Source Auction (10,000 kW)

Bid year	Category	Cleared capacity (10,000 kW)
FY2023 bids	Batteries and pumped hydro	166.9
FY2024 bids	3 to 6 hours	96.1
FY2024 bids	6 hours or more	76.9
FY2025 bids	Lithium-ion	81.9
FY2025 bids	Non-lithium	88.6

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Source: Clearing results and capacity assurance contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in section 15). As published.

18. Assumptions and terms

- The figures are our own tally of the values published in the sources. We have not added up several months or converted anything to annual figures. We do not show revenue amounts for any single battery storage plant.
- Months covered: wholesale electricity (JEPX) covers October 2026 (31 days), and the balancing market (EPRX) covers preliminary figures for September 2026 (delivery dates up to the 29th). The balancing market will cover a different month until the published figures for October 2026 are in our tally.
- Average prices are simple averages of the 30-minute slots (not weighted by trading volume). Daytime means the slots from 08:00 to 16:00, and evening means the slots from 16:00 to 20:00.
- Price gap: we rank the day's 30-minute slots by price and take the average of the highest 8 slots (4 hours) minus the average of the lowest 8 slots (4 slots for 2 hours). It does not consider whether the cheap hours come before or after the expensive hours, electricity lost in charging and discharging, or fees, taxes and transmission charges. It does not show actual revenue.
- Balancing market: the nationwide figures (total column) of the EPRX preliminary figures. Volume requested is by grid operator (TSO); bid volume, winning volume and average winning price are by the area where the power source is located. Volumes are averages per 30-minute slot. The average winning price is weighted by winning volume. Bid ratio = bid volume ÷ volume requested; clearing rate = winning volume ÷ bid volume. For the 3-hour blocks up to March 13, 2026, we counted each block as six 30-minute slots.
- Range of winning prices: the thin line runs, among the slots with winning bids in the month, from the lowest winning bid (lowest winning price) to the highest winning bid (highest winning price) (published figures by the area where the power source is located). The published figures for Primary reserve, Secondary reserve 1 and 2 and Tertiary reserve 1 also include composite-product bids (one price for several products bundled together), so bids above 10 yen that are explained by a composite-product bid in the same slot and area are left out of the thin line. The thick band is the range from the bottom 10% to 90% when the slots' average winning prices are sorted from lowest. The dot is the average weighted by winning volume.
- Clearing rate by grid area and time of day: the total winning volume ÷ the total bid volume of the slots at the same time of day in the month (by the area where the power source is located).
- The time from instruction to change in output and the duration for each balancing market product are the FY2026 specifications.
- High-price slots for Tertiary reserve 2: slots where the nationwide average winning price was 10 yen/kW per 30 min or more. The top 1% means the top 1% of slots ranked by revenue (average winning price × winning volume). The maximum winning prices are the published values by the area where the power source is located.

Sources (OCCTO, capacity market clearing results): delivery FY2024 https://www.occto.or.jp/iinkai/youryou/kentoukai/2020/files/youryou_kentoukai_27_05.pdf / delivery FY2025 https://www.occto.or.jp/iinkai/youryou/kentoukai/2021/files/youryou_kentoukai_35_03_1.pdf / delivery FY2026 https://www.occto.or.jp/iinkai/youryou/kentoukai/2022/files/youryou_kentoukai_43_03.pdf / delivery FY2027 https://www.occto.or.jp/market-board/market/oshirase/2023/files/240124_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2027.pdf / delivery FY2028 https://www.occto.or.jp/market-board/market/oshirase/2024/files/250129_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2028.pdf / delivery FY2029 https://www.occto.or.jp/assets/news/capacity-market/260120_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2029.pdf

Sources (OCCTO, Long-Term Decarbonization Power Source Auction): FY2023 bids https://www.occto.or.jp/assets/260605_auction_keiyakujyoukyou_kouhyou.pdf / FY2024 bids https://www.occto.or.jp/market-board/market/jitsujukyukanren/files/longauction_keiyakukekka_2024.pdf / FY2025 bids https://www.occto.or.jp/assets/various/capacity-market/jitsujukyukanren/yoryokakuhokeiyaku/longauction_keiyakukekka_2025.pdf

- Capacity market clearing results are published in a different year for each delivery fiscal year (FY2024 in 2020, FY2029 in January 2026). The number of battery projects in the Long-Term Decarbonization Power Source Auction is not shown, because it is not in the published summary tables.

Sources: Japan Electric Power Exchange (JEPX) spot market trading results / Electric Power Reserve Exchange (EPRX) trading results (preliminary) and trading results by power source type / Organization for Cross-regional Coordination of Transmission Operators (OCCTO) capacity market main auction clearing results and Long-Term Decarbonization Power Source Auction capacity assurance contract documents / our tally of the grid connection study responses we received / our DealRoom records.

Terms: system price (System (nationwide)) = the wholesale electricity price when all of Japan is treated as one market. Area price = the price that splits by area depending on how congested the transmission lines are. The grid = the power companies' transmission network. Grid connection = connecting to the transmission network. MW = the size of electric power (1 MW = 1,000 kW). MWh = the amount of electricity that can be stored. yen/kWh = the price per 1 kWh of electricity.

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